

User manual for SPIFFI code

1. Runtime environment

The code has been tested in MATLAB 2022, 2023 and 2024.

2. How to run this code:

(1) Install Signal Processing Toolbox and Image Processing Toolbox in MATLAB

(2) Run the SPIFFI_main_X.m

(3) Demo results can be found in the data folder

3. Reconstruction Workflow

The following steps outline the **MAIN processing pipeline** of the SPIFFI.

Users can customize the reconstruction steps based on the reference workflow:

(1) **Data Input:** Load data containing 4 polarization channels (0° , 90° , 45° , 135°).

(2) **Virtual Resampling and Shuffling:** Apply *pixelSplit* function to each channel to create sub-pixel variations. Apply *randperm* (shuffling) function to simulate the signal fluctuation.

(3) **Core Reconstruction:** Execute SPIFFI_functions using defined cumulant orders and PSF models.

(4) **Linearization and Visualization:** Apply power-law correction based on the chosen reconstruction order to restore image linearity.

4. Parameter Definitions and Suggestions (cfg structure)

(1) Optical system parameters

Parameter	Description	Note
pxSize	Pixel size of the camera at sample plane (unit: meters)	Calibrate your pixel size using standard samples instead of directly using theoretical calculations.
lambda	Emission wavelength of the fluorophore (meters)	-
NA	Numerical Aperture of the objective lens	-
PSFshrink	PSF contraction factor under polarization	1.2 - 1.5 (Default: 1.3). This value is derived from the ratio of the major axis length to the minor axis length of the polarized PSF.

(2) Reconstruction parameters

Parameter	Description	Note
M	Magnification factor for the super-resolution image	≥ 2 , The reconstructed resolution should satisfy the Nyquist sampling criterion.
order	Correlation order for resolution enhancement	3th is the best
ifAC	Autocorrelation (True) or Cumulant-based analysis (False)	ifAC=truth is slow but less artifact, which calculates all delays of the fluctuation data, commonly used for single-shot reconstruction. However, when utilizing continuous frame data (ifT=true) or temporal fluctuations (two-stage temporal reconstruction, ifAC2=true), the operation with ifAC=false is faster.
ACscale	Scaling factor for the fluctuation analysis	When ACscale<1, SPIFFI preserves a portion of the direct current component in the signal to maintain the main body of the target structure, making it suitable for single-shot reconstruction. When ACscale=1, the algorithm focuses on fluctuating components, which is the standard SOFI-AC, making it suitable for two-stage reconstruction combined with temporal fluctuations.

ifAC2	Enable two-stage reconstruction (spatial + temporal)	(ifAC2=false) for the single-shot; (ifAC2=true) for the two-stage; Please set ACscale=1 for the ifAC2=true;
ifPreDeconv	Enable Richardson-Lucy deconvolution before fluctuation analysis	This could enhance the performance of the reconstruction
deconvPre	Number of iterations for pre-deconvolution	5-15 iterations, stop iteration before artifacts are generated.
deconvPost	Number of iterations for post-reconstruction sharpening	2-5 iterations, enhance image contrast
ifbkg	Enable rolling ball background subtraction	Adjust the background removal effect using the radius of the ball.
ifT	Enable time-related continuity analysis	Using a small number of frames to improve the signal-to-noise ratio of the reconstruction
tframes	Number of frames used in continuity analysis	e.g. tframes=1 when ifT=true. Reconstruction is performed by combining the previous and next frames (a total of three frames).
bkgRatio	Remove the background residuals in linearization	<0.1

Most parameters in SPIFFI do not require additional adjustment after correct calibration; typically, only **ifAC**, **ACscale**, **deconvPre** and **ifbkg** need to be adjusted based on the signal-to-noise ratio.

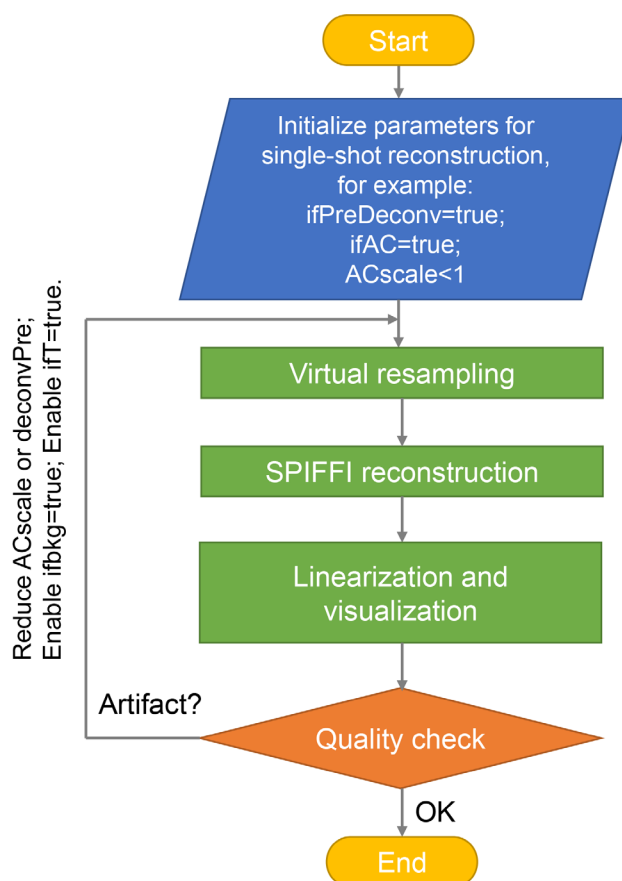
5. Demonstrations

We provide three demos to show how to customize the reconstruction process:

(1) Single-shot demo (microtubules)

Execute the SPIFFI_main_singleshot.m file to reconstruct microtubule image. For single-frame super-resolution, it is recommended to use ifAC=true, i.e., full-delay autocorrelation, to improve image fidelity. In

addition, please note that ACscale should be less than 1 (usually 0.7 is sufficient) to ensure the integrity of the target structure. For a high SNR raw data, background subtraction is not necessary (ifbkg=false). But for a SNR low data, background subtraction is recommended (ifbkg=true).

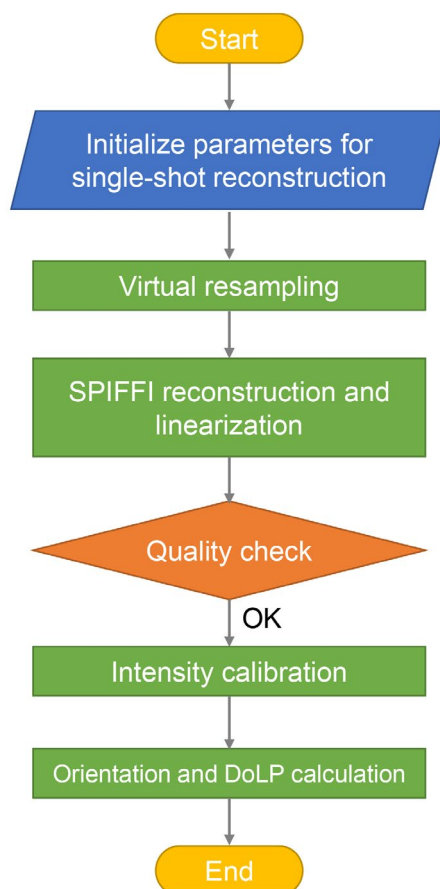


Workflow of the Single-shot reconstruction

(2) Orientation demo (GUV)

Execute the SPIFFI_main_orientation.m file to reconstruct GUV image and orientations. The SPIFFI reconstruction workflow is identical to that of single-shot super-resolution, but intensity calibration and background removal **must** be performed before calculating molecular orientation. Both intensity imbalance between channels and fluorescence background bias

can lead to errors in dipole orientation calculations.

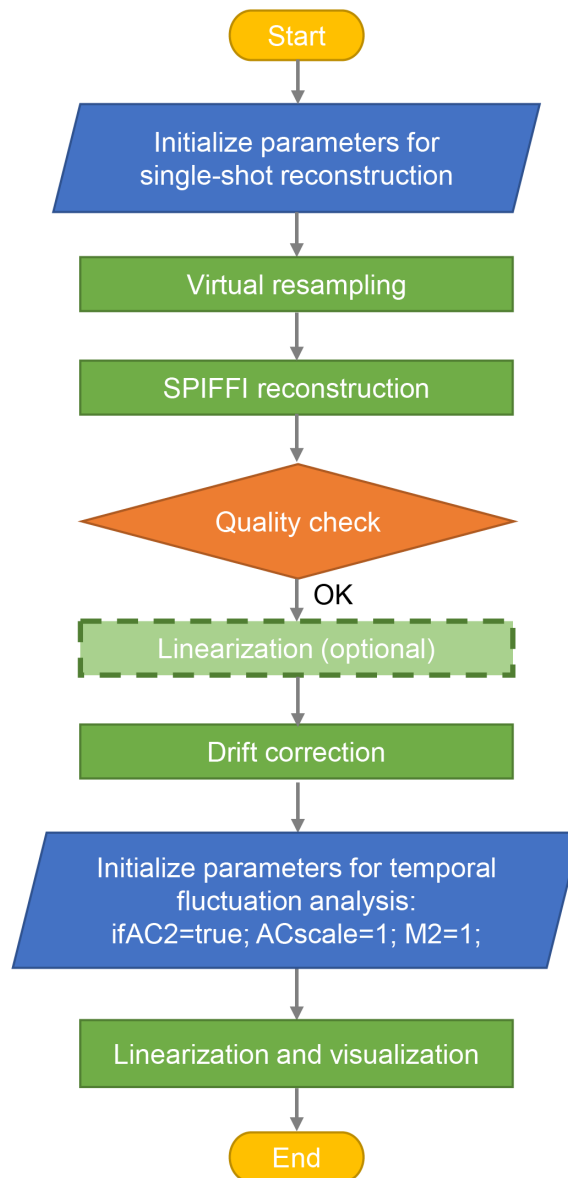


Workflow of the orientational calculation

(3) Two-stage demo (80 nm nanorulers)

Execute the SPIFFI_main_twostage.m file to reconstruct the nanoruler image. The SPIFFI reconstruction workflow is identical to that of single-shot super-resolution. However, linearization processing of each frame after SPIFFI processing is an optional step, because the linearization process (n -th root operation) aligns the image brightness more closely with the physical density distribution of molecules, but compresses the fluctuation contrast of fluorescence. Furthermore, drift correction is particularly crucial for reconstructing long-term sequence images. Finally,

during temporal fluctuation analysis, set ifAC2=true, where the number 2 indicates reconstruction of the second stage. Set ACscale=1 to force the algorithm to focus on the fluctuation signal. Upsampling is typically not performed in the second stage (SOFI-AC). Therefore, the magnification (upsampling) factor in the first stage (SPIFFI single-shot) should ensure that the pixel size of the final image (SPIFFI+SOFI) satisfy the Nyquist sampling criterion.



Workflow of the two-stage reconstruction